STAT 4830 - SPRING 2026

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Optimizing Hyperparameter Tuning Methods for *Evolutionary Strategies*

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Problem Statement

Problem Statement

- **Goal:**
  1. Set up Hugging Face and Prime Intellect GPU (and potentially vLLM) to run experiments on laptop
  2. Run a grid-search style hyperparameter search to find an ideal **learning rate** and **variance scale**
  3. With those ideal hyperparameter set, run ablation studies on the impact of **Population** (1-32), **Antithetic perturbation** (one-sided vs. two-sided), **Noise type** (Gaussian vs. Rademacher), **Normalization** (on vs. off), **Top-k sampling** (on vs. off)

02
Technical Approach

Technical Approach

- **Model**: OPT-350M (330M params) in bfloat16 on NVIDIA A40 GPU via Prime Intellect
- **Tasks**: RTE (277 val examples) and BoolQ (500 val examples); reward = +1 correct first-token prediction, -1 otherwise
    - RTE turned out to be non-variant across different configurations so moved on to BoolQ later

- **Algorithm**: Two-sided antithetic ES — for each seed, evaluate $\theta+\sigma\varepsilon$ and $\theta-\sigma\varepsilon$, compute advantage $a_k = r^+ - r^-$, z-score normalize, apply gradient update
- **Memory efficiency**: Seed-based in-place perturbation (MeZO-style) — noise reconstructed on-the-fly from integer seed, no stored noise vectors, no model copies (~50MB peak overhead vs ~1.2GB naive)
- **Two-phase protocol**: Calibration first (4×4 $\sigma\times\alpha$ grid, 5 seeds each, 20 iters) → select best HPs → ablation blocks at fixed compute budget
- **Fairness control**: All ablation variants matched on training forward passes (not wall-clock, not iterations) — one-sided ES runs 2× iterations to consume the same budget as two-sided
- **Ablation blocks**: Population size ($N\in\{1,4,8,16,32\}$), antithetic vs. one-sided, Gaussian vs. Rademacher noise, normalization on/off, top-k selection (all/top-4/top-8)
- 5 seeds per variant for reliable mean and std estimates across all blocks

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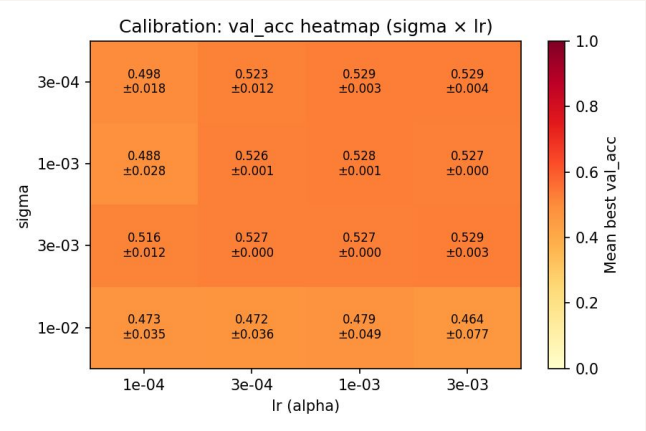
Results

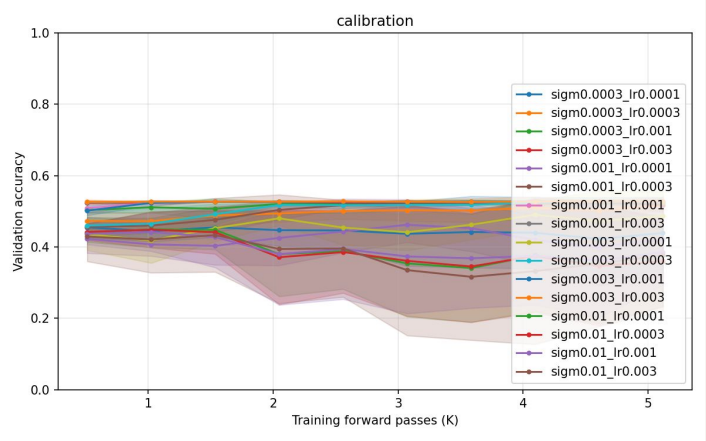
Results - Goal 1

- **Successfully hooked up GPU and HF**
  - Primarily used A40 48GB, which was enough to run models up to 7B param count
  - Could be scalable to further studies by using a GPU with higher VRAM & latest version
  - Haven't set up vLLM yet since we yet need multiple-GPU system. We will set up later accordingly on demand

Results - Goal 2

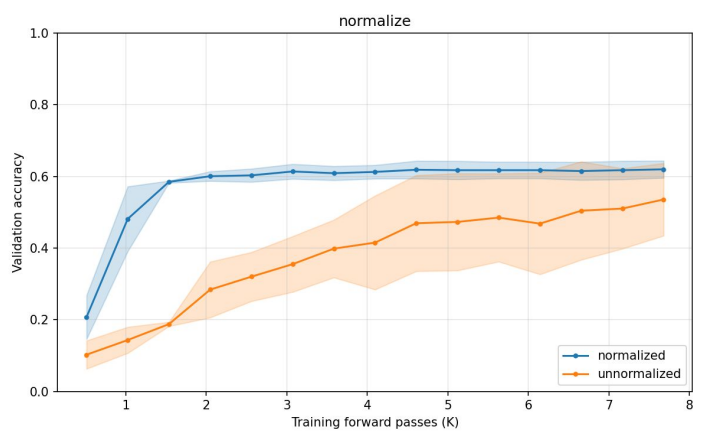
- **Hyperparameter Sensitivity**
  - Best configs: σ=3e-4 with α=1e-3 or α=3e-3 (both 0.529) — consistent with week 8 findings
- **Clear patterns**:
  - σ=1e-2 is the hard failure zone — all configs 0.464–0.479, highest variance (±0.077)
  - α=1e-4 is unreliable across all σ — lowest means, highest std
  - The safe zone is σ∈{3e-4, 1e-3, 3e-3} × α∈{3e-4, 1e-3, 3e-3} — all cluster tightly around 0.523–0.529

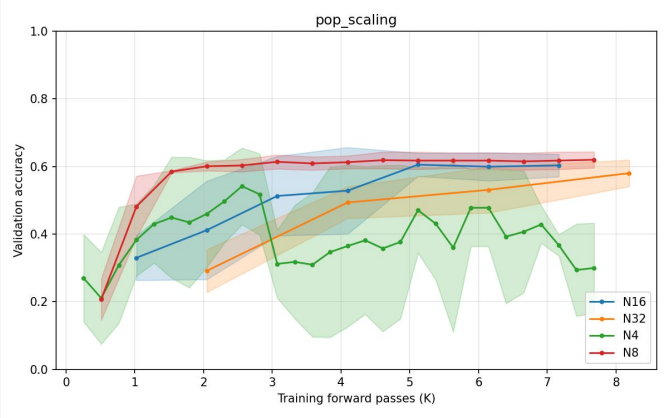




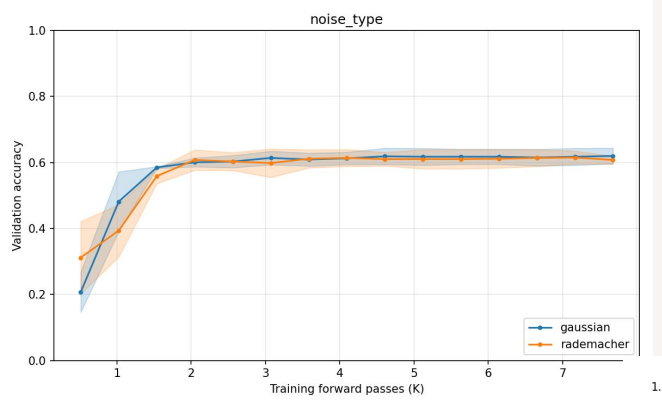
Results - Goal 3

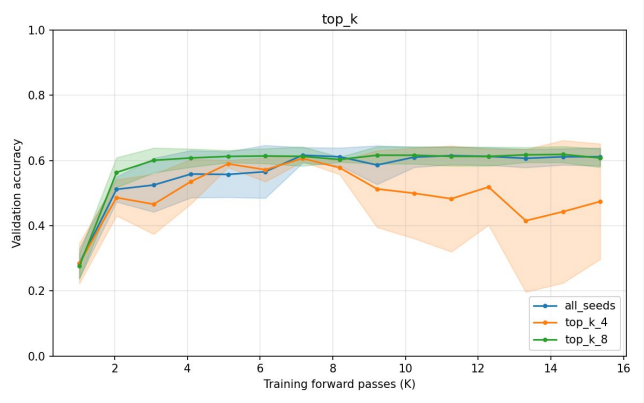
- **One_vs_two:** No meaningful difference
- **Noise_type:** Rademacher converges slightly faster early (~1K passes) but both reach the same ~0.62 plateau. No practical advantage for either on BoolQ.
- **Normalize:** Clearest result of all ablations. Normalized (blue) converges fast to 0.62 by 2K passes with low variance. Unnormalized (orange) barely reaches 0.54 by 7.5K passes with massive variance. Z-score normalization is critical.
- **Top_k**: top_k=4 is unstable and degrades over time, dropping below 0.5 by 14K passes. top_k=8 and all_seeds both reach ~0.61 and stay stable. Top-k selection hurts when k is too small — filtering too aggressively introduces bias**.**
- **Pop_scaling:** Messy with high variance, especially N4 (green). N32 looks slightly better but hard to tell cleanly. The general trend suggests larger N is more stable, but the differences are small after convergence. Need more extensive studies

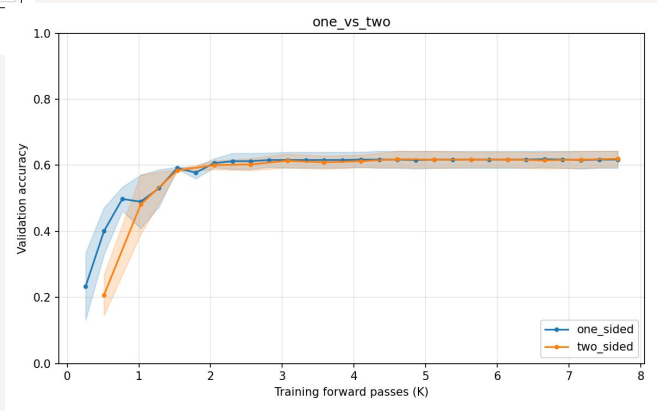




Results - Goal 3 (Plots cont'd)







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Limitations

Limitations

- **Still confined to a single model (OPT-350M) and single task (RTE)**
  - Need to verify on larger scale (billions) and multiple task types (classification / multiple-choice / generation)
- **N=8 is theoretically very small for d≈330M, though empirically sufficient here**
  - Need more study about the tradeoff between N (population) vs. T (step count), especially since many previous papers used N=1 with thousands of steps
- **Still haven't employed SoTA techniques such as sparse update, hessian preconditioner, low-rank tricks, etc.**
  - Plan to implement next week

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Next Steps

Next Steps

- **Further Ablation Studies**
  - Scale Up
    - Size: OPT-1.3B / 7B
    - Task: CB / WSC / etc.
  - Incorporate Batch Size (B)?
  - Subspace? (LoRA, etc.)
- **Compare SOTA ZO methods (Sparse-MeZO, HiZOO, ESSA, etc.) under same forward-pass budget**
  - Goal: dissect where the tradeoff exists and which one fits in which environment.

THANK YOU

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